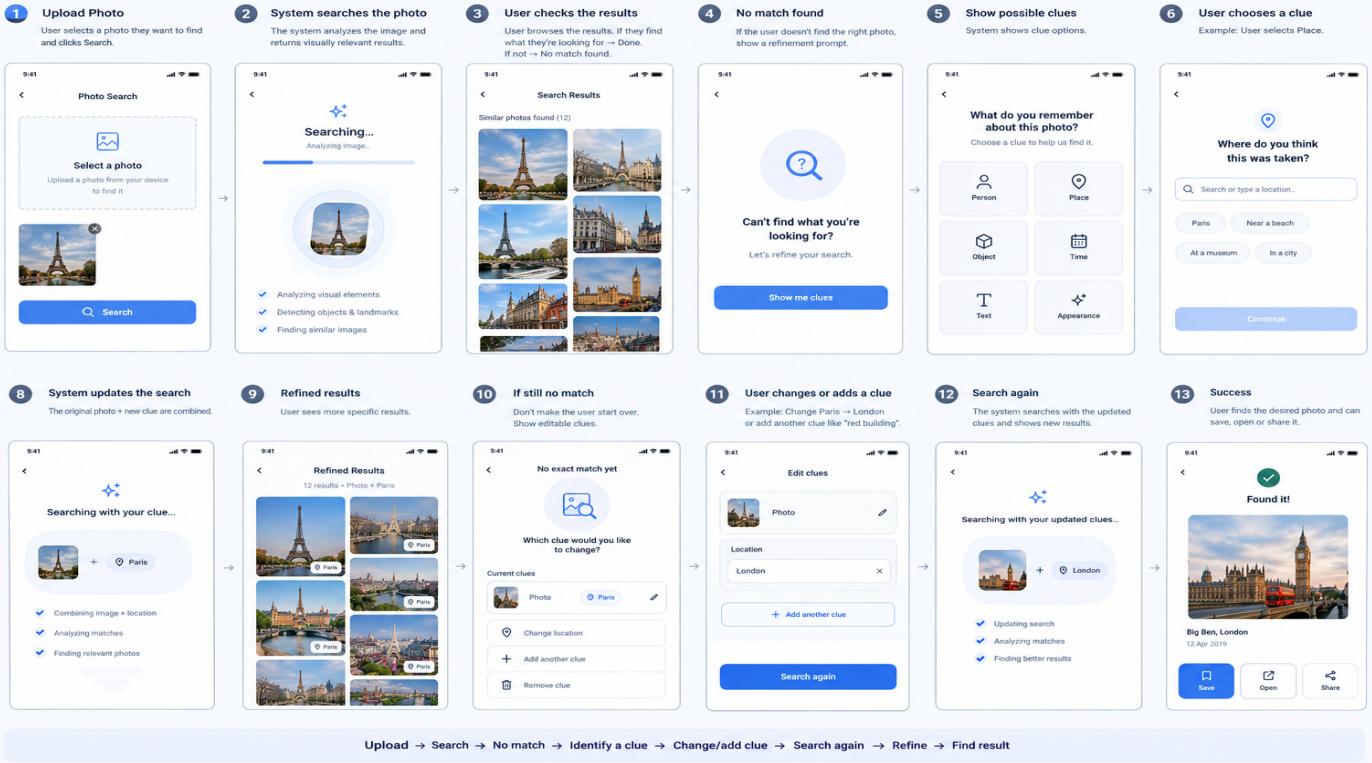
VISUAL CLUE SUGGESTIONS

A guided AI refinement layer for photo retrieval



Product goal

Help users retrieve vaguely remembered photos without forcing them to invent a new search query.

PM thesis

When the first search fails, the product should provide the next useful clue—not another blank search box.

Prototype: photo-hint-helper.lovable.app | Presentation structure inspired by strong product-case storytelling: problem evidence opportunity solution MVP metrics experiment.

1. Executive summary

The retrieval gap is not only search quality. It is the user's ability to know what to try next when the first search fails.

User problem

People remember the moment, not exact searchable terms.

Observed friction

Failed searches lead to query guessing + scrolling.

Opportunity

Turn memory into a guided sequence of searchable clues.

1.5B+

monthly Google Photos users

370M+

monthly photo-search users

9T+

photos + videos stored

Market context: Google reported these figures on May 28, 2025. The scale makes retrieval efficiency a meaningful product surface.

2. Problem definition

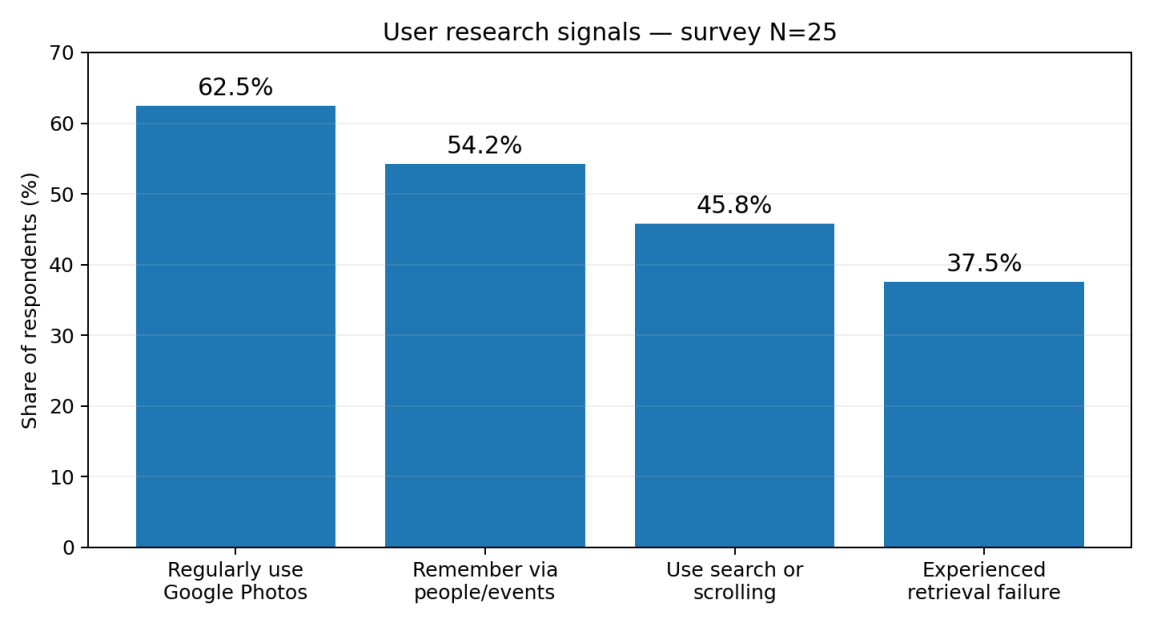
Core problem: users remember the experience, but not necessarily the searchable information required to retrieve it.

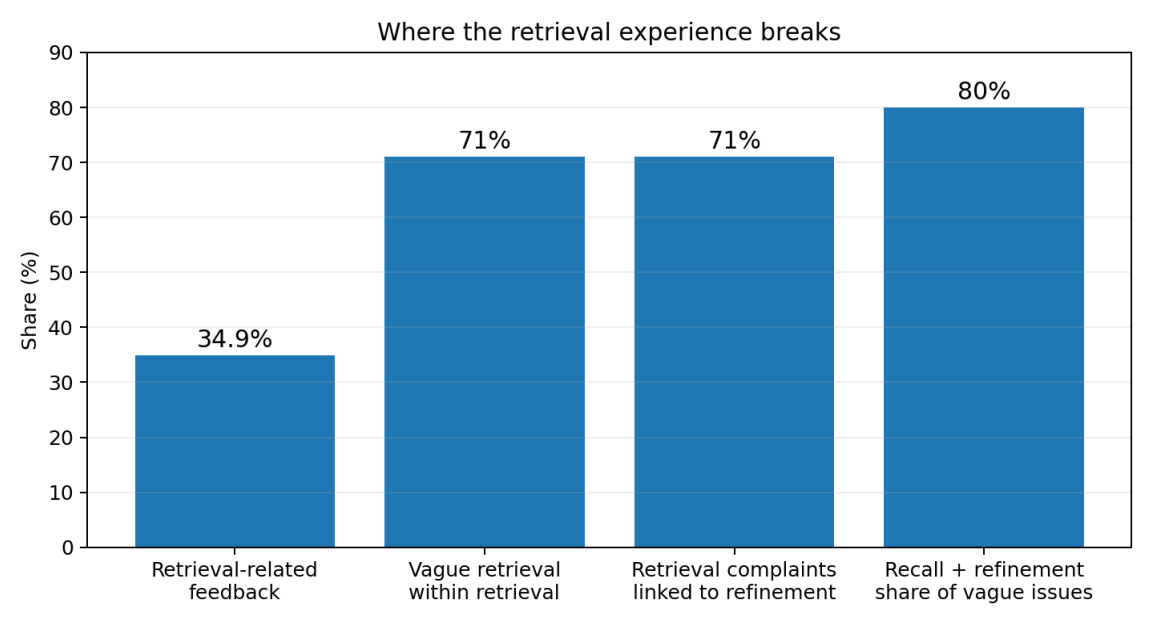
What users have	What search expects	Where the gap appears
People • place • event • object • approximate time	Names • dates • keywords • structured filters	The first query is incomplete, broad or wrong
A mental picture or story	Textual / semantic signals the system can match	User does not know which clue to change

Product implication: a failed search should become a guided refinement moment instead of a dead end.

3. Evidence: what the research says

Primary research synthesis from the project: survey N=25 plus review analysis of 456 feedback items.





Interpretation: retrieval failure is a recurring behavior, and refinement failure is the product-design opportunity. These are project research signals, not externally audited market statistics.

4. Competitive landscape & whitespace

The category already supports semantic search, faces, places, dates, OCR and AI-assisted search. The whitespace is the recovery journey after an unsuccessful search.

Product	Strength	Observed gap / opportunity
Google Photos	Natural-language search; people, places, objects, dates; Ask Photos	Guide users more explicitly when a vague search fails
Apple Photos	Natural-language + people / objects / places	Make memory-to-search refinement more explicit
Amazon Photos	AI recognition + keyword/context search	Improve retrieval from incomplete clues

Immich	Semantic search + faces + OCR + metadata	Add adaptive guidance for users who do not know the next filter

Whitespace: existing systems help users express a search; this concept helps users recover when expression is incomplete.

5. Opportunity & product hypothesis

Opportunity statement

Build an adaptive retrieval experience that converts vague memory into increasingly useful search clues.

Core hypothesis

If users can select a visual clue after a failed search, they can refine vague memories with less effort and improve successful retrieval.

Trigger	Intervention	Behavior	Outcome
Initial search fails	Show clue choices	User adds / edits clue	More relevant results

Design principle: recognition over recall — let users choose what they remember instead of forcing them to formulate the perfect query.

6. Solution: Visual Clue Suggestions

After a weak result set, the interface asks: “What do you remember about this photo?”

Person Who was there?	Place Where was it?	Time When was it?
Object What was visible?	T Text Any words / signs?	Appearance Color / visual detail?

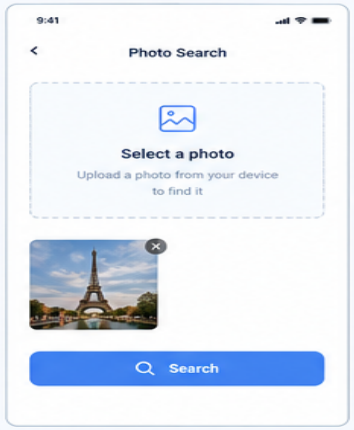
Example: Photo + “Paris” Photo + “Paris” + “red building” refined results. Existing clues remain editable.

7. MVP user journey

The MVP intentionally keeps the interaction loop narrow: search recover refine retrieve.

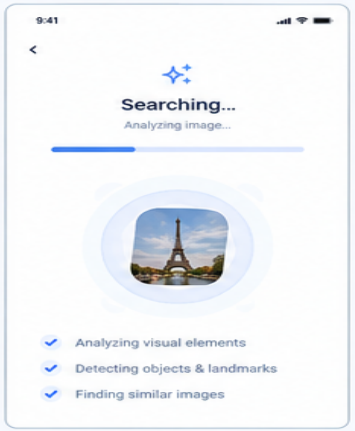
1 Upload Photo

User selects a photo they want to find and clicks Search.



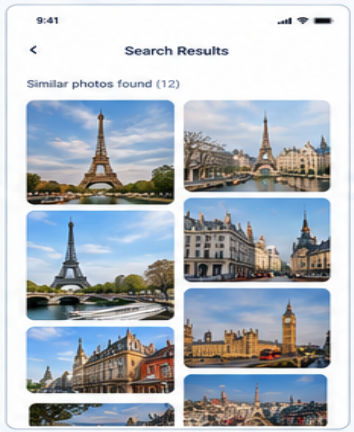
2 System searches the photo

The system analyzes the image and returns visually relevant results.



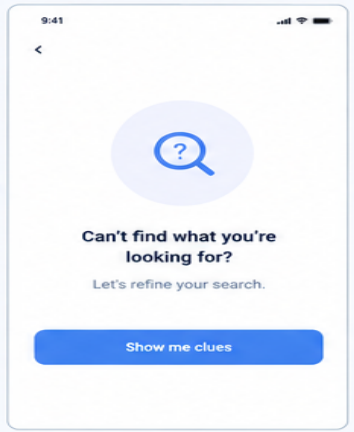
3 User checks the results

User browses the results. If they find what they're looking for → Done.
If not → No match found.



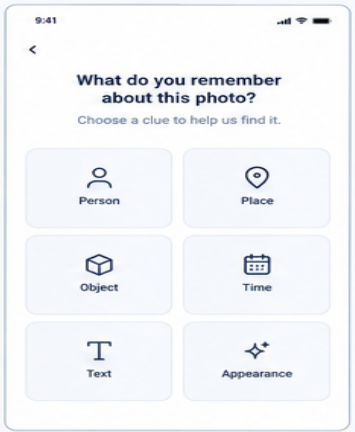
4 No match found

If the user doesn't find the right photo, show a refinement prompt.



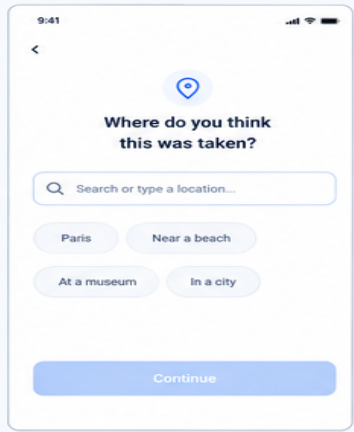
5 Show possible clues

System shows clue options.



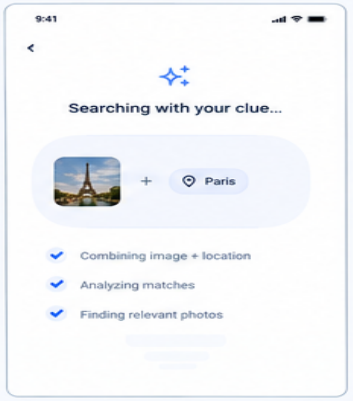
6 User chooses a clue

Example: User selects Place.



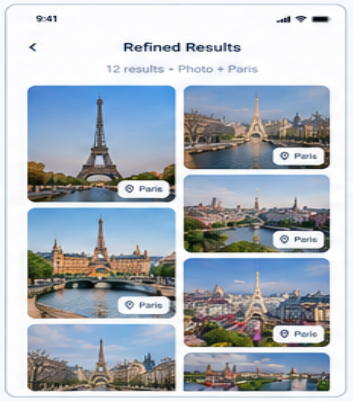
8 System updates the search

The original photo + new clue are combined.



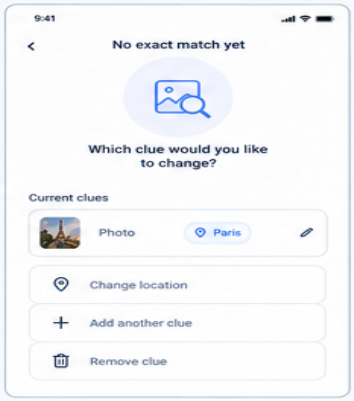
9 Refined results

User sees more specific results.



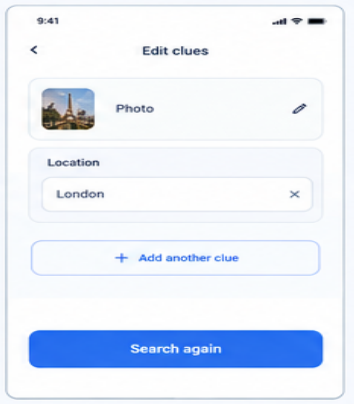
10 If still no match

Don't make the user start over.
Show editable clues.



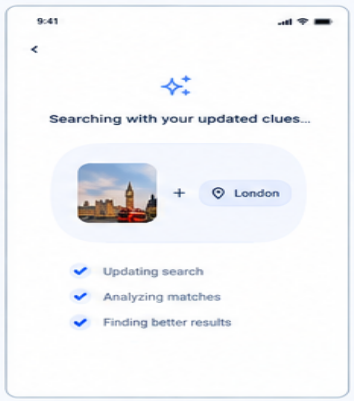
11 User changes or adds a clue

Example: Change Paris → London or add another clue like "red building".



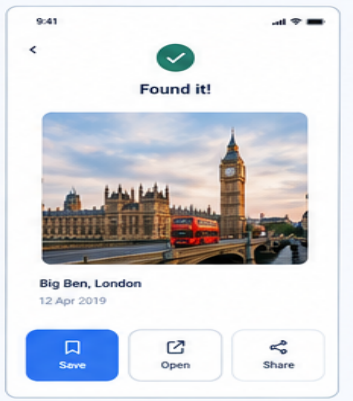
12 Search again

The system searches with the updated clues and shows new results.



13 Success

User finds the desired photo and can save, open or share it.



Upload → Search → No match → Identify a clue → Change/add clue → Search again → Refine → Find result

8. MVP scope & product behavior

Must have

Upload photo
Initial results
No-match state
Clue selection

Must have

Focused follow-up
Editable clues
Refined results
Save / Open / Share

Not in MVP

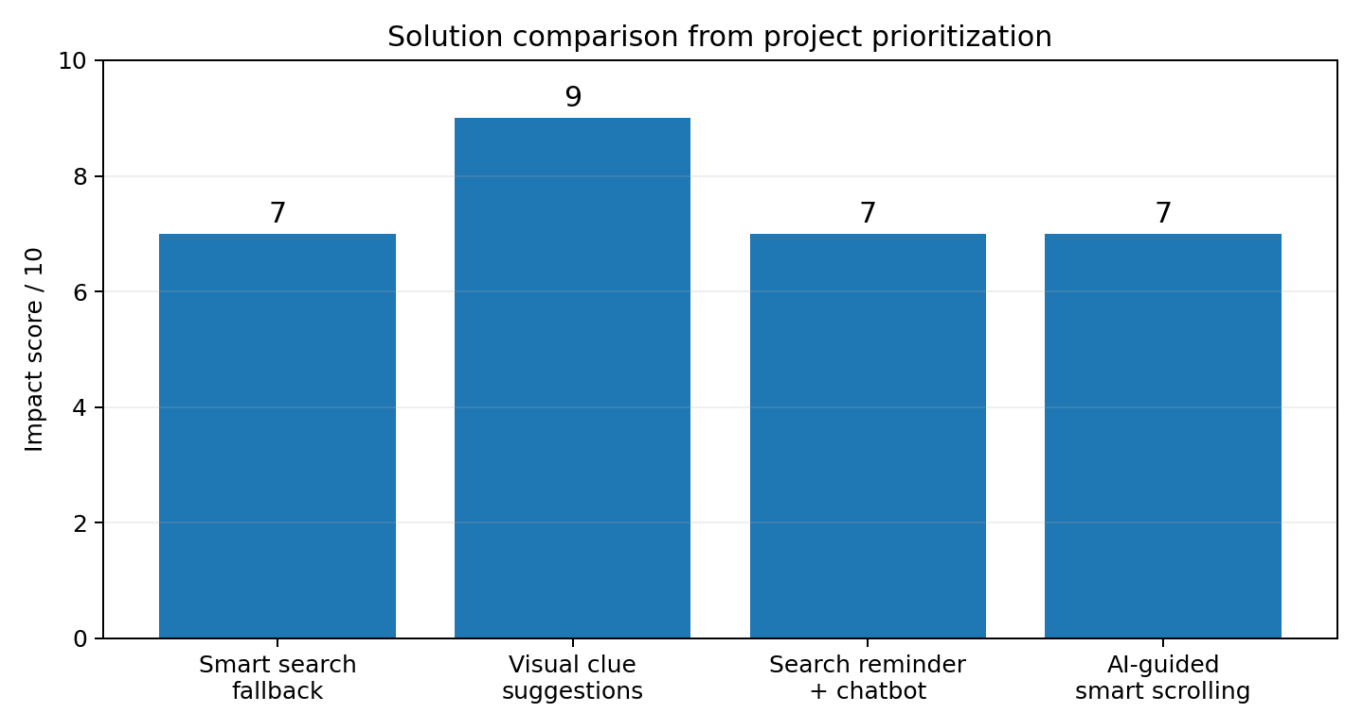
Full semantic ranking
Personalized clue ranking
Long-term memory
Advanced multimodal agents

MVP principle: prove the refinement behavior before investing in a full retrieval engine.

Prototype link: photo-hint-helper.lovable.app

9. Prioritization: why this solution

The project compared four solution directions using impact, confidence, effort and fit.



Visual Clue Suggestions directly addresses the refinement gap: it changes the user's next action from “guess another query” to “choose a clue.”

Source: project prioritization framework. Scores are internal decision-support estimates, not measured user outcomes.

10. Success metrics & experiment design

North-star outcome: successful retrieval of a vaguely remembered photo.

Metric	Definition	Why it matters
Guided refinement success	Refined searches that lead to a relevant result	Directly validates the core hypothesis
Time to find	Time from first search to successful retrieval	Measures efficiency
Search attempts	Number of search/refinement cycles per task	Detects trial-and-error reduction
Result relevance	User-rated relevance of the refined result set	Protects retrieval quality

A/B test: failed-search users see either (A) a standard retry/search box or (B) Visual Clue Suggestions. Compare retrieval success, time-to-find, attempts and abandonment.

11. Risks, rollout & next steps

Risk	Mitigation
Wrong or irrelevant clue suggestions	Rank clues using detected entities + confidence; always allow manual entry.
Too many choices increase cognitive load	Start with 4–6 high-value clues; personalize the order later.
Refinement improves relevance but not exact recall	Track both relevance and successful retrieval; do not optimize only for click-through.
Users may abandon after multiple failures	Keep clues editable, preserve context, and show progress toward narrowing.

Phase 1 — Prototype: validate clue selection and follow-up UX.

Phase 2 — Pilot: instrument retrieval success and compare against baseline.

Phase 3 — Scale: personalize clue ranking using retrieval signals and user feedback.

12. PM takeaway

The product opportunity is not simply “better search.”

It is better recovery from search failure.

A strong retrieval experience should turn vague memory into a sequence of useful clues, preserve context between attempts, and make the next action obvious.

Product narrative: Problem Evidence Opportunity Guided solution MVP Metrics Experiment Scale

References: Google Photos official product update (May 28, 2025); Spotify Investor Day 2026 product-strategy framing; project research synthesis and prototype.
Google Photos source: blog.google/products-and-platforms/products/photos/google-photos-10-years-tips-tricks/
Spotify source: newsroom.spotify.com/2026-05-21/investor-day-recap/